

Oluwatobi Melvyn Mayungbo

Data and Machine Learning Engineer

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SUMMARY

Data and machine learning engineer with a B.Sc. in Statistics. Builds tested data pipelines in Python and SQL with Prefect, dbt and DuckDB, a payment reconciliation pipeline with audit trails, and a PostgreSQL ledger API with 138 tests. Works with Git, CI/CD and Docker.

SKILLS

Engineering and MLOps: Python, SQL, dbt-core, Prefect, DuckDB, PostgreSQL, Git, CI/CD (GitHub Actions), Docker, Redis, FastAPI, GCP Cloud Run, AWS ECS Fargate, Railway, MLflow, Evidently, Pytest, Streamlit, React

Payments and Risk: Payment Reconciliation, Double-Entry Ledgers and Idempotency, Transaction Fraud Detection, Credit Risk Decisioning, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Engineered a multi-stage reconciliation pipeline using DuckDB and FastAPI, with temporal join logic and tolerance thresholds to handle asynchronous processor webhooks and bank settlement files.
- Containerised and deployed the full service on Railway with structured log collection, automated integration testing and persistent transaction state tracking.

Production-Grade Double-Entry Ledger API | [GitHub](#)

Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.

- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER**

May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action